

Introduction to Cloud Networking

Task 1

Log in to your AWS account, open the VPC dashboard, and create a new VPC named InstaVibeVPC with a custom IPv4 CIDR block (e.g., 10.0.0.0/16).

Steps followed:

1. Logged in to the AWS Management Console and searched for "VPC" in the top search bar to open the VPC Dashboard.
2. Clicked on "Your VPCs" in the left-hand menu, then clicked the "Create VPC" button.
3. Selected "VPC only" as the creation option (instead of the full VPC-and-more wizard) for a manual, step-by-step setup.
4. Entered the name tag as "InstaVibeVPC".
5. Set the IPv4 CIDR block to a custom range, 10.0.0.0/16, to allow enough IP addresses for future subnets.
6. Left IPv6 CIDR block and tenancy at their default settings, then clicked "Create VPC".
7. Confirmed on the "Your VPCs" page that InstaVibeVPC was listed with the state "Available" and CIDR 10.0.0.0/16.

Task 2

Inside your InstaVibeVPC, create two subnets: one public subnet (e.g., 10.0.1.0/24) for hosting a web server and one private subnet (e.g., 10.0.2.0/24) for backend services.

Steps followed:

1. In the VPC Dashboard, clicked on "Subnets" in the left-hand menu, then clicked "Create subnet".
2. Selected "InstaVibeVPC" as the VPC ID for both subnets.
3. For the first subnet, set the subnet name to "PublicSubnet-Zomato", chose an Availability Zone, and set the IPv4 CIDR block to 10.0.1.0/24.
4. Clicked "Add new subnet" and created a second subnet named "PrivateSubnet-Zomato", using the same VPC and a different Availability Zone or the same one, with the IPv4 CIDR block set to 10.0.2.0/24.
5. Clicked "Create subnet" to finalise both subnets in one step.
6. Verified on the Subnets page that both "PublicSubnet-Zomato" (10.0.1.0/24) and "PrivateSubnet-Zomato" (10.0.2.0/24) were listed under InstaVibeVPC.

Task 3

Configure an Internet Gateway and attach it to your InstaVibeVPC so that only the public subnet can access the internet, while the private subnet cannot.

Steps followed:

1. In the VPC Dashboard, clicked on "Internet Gateways" in the left-hand menu, then clicked "Create internet gateway".
2. Named it "InstaVibeVPC-IGW" and clicked "Create internet gateway".
3. Selected the newly created internet gateway, clicked "Actions", and chose "Attach to VPC".
4. Selected "InstaVibeVPC" from the list and confirmed the attachment, so the internet gateway's state changed to "Attached".
5. Made sure the internet gateway itself was not attached or referenced anywhere in the private subnet's route table, so that only the public subnet would be able to use it for outbound/inbound internet traffic (configured fully in Task 4).

Task 4

Set up route tables so that your public subnet routes traffic to the Internet Gateway, but your private subnet does not. Verify this by checking the route table associations.

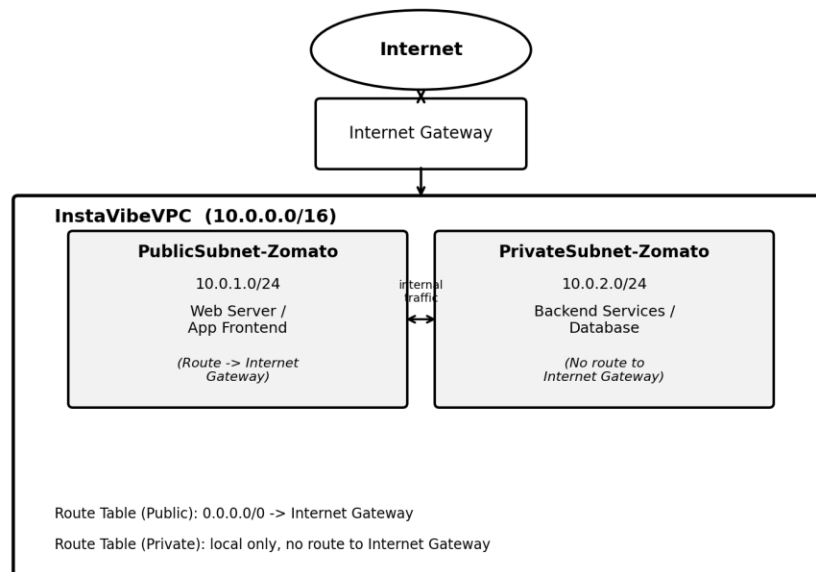
Steps followed:

1. In the VPC Dashboard, clicked on "Route Tables" and created a new route table named "PublicRouteTable-Zomato", associated with InstaVibeVPC.
2. Selected "PublicRouteTable-Zomato", opened the "Routes" tab, clicked "Edit routes", and added a route with destination 0.0.0.0/0 pointing to the InstaVibeVPC-IGW internet gateway, then saved it.
3. Opened the "Subnet associations" tab of PublicRouteTable-Zomato, clicked "Edit subnet associations", and selected "PublicSubnet-Zomato", then saved.
4. Checked the private subnet's route table (the VPC's default/main route table) and confirmed it only contained the default "local" route for 10.0.0.0/16, with no route to the internet gateway.
5. Made sure "PrivateSubnet-Zomato" remained associated with this private/main route table and was not linked to PublicRouteTable-Zomato.
6. Verified the setup on the "Route Tables" page under "Subnet associations" for each route table — PublicSubnet-Zomato appeared only under PublicRouteTable-Zomato (with the internet gateway route), while PrivateSubnet-Zomato appeared only under the private route table (with no internet gateway route), confirming the public subnet has internet access and the private subnet does not.

Task 5

Draw a simple diagram (hand-drawn or using any online tool) showing your VPC, public subnet, private subnet, and Internet Gateway, and briefly explain in 3-4 lines how this setup is similar to how apps like Zomato separate their public user interface from their private databases.

Diagram:



Explanation:

This VPC setup mirrors how a real app like Zomato is structured behind the scenes. The public subnet, which connects to the internet gateway, is similar to Zomato's web and app servers that users directly interact with when browsing restaurants or placing orders. The private subnet, which has no route to the internet gateway, is similar to Zomato's backend databases that store sensitive information such as user details, payment records, and order history — these are never exposed directly to the internet. Just like in this VPC, where only the public subnet can be reached from outside, Zomato's actual database servers can only be accessed through its own application layer, keeping sensitive data protected from direct public access while still letting users use the app normally.